

Endocrine physiology, pathophysiology and pharmacology: Pituitary and Hypothalamus

Integrated Biomedical and Clinical Sciences OMFS 513

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Mission: To advance the science and art of health through education, service, scholarship and social accountability

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Learning Objectives

- Describe the hormone function
- Describe the normal function and structure of the pituitary gland
- Describe select pathological conditions associated with the pituitary gland and their management.

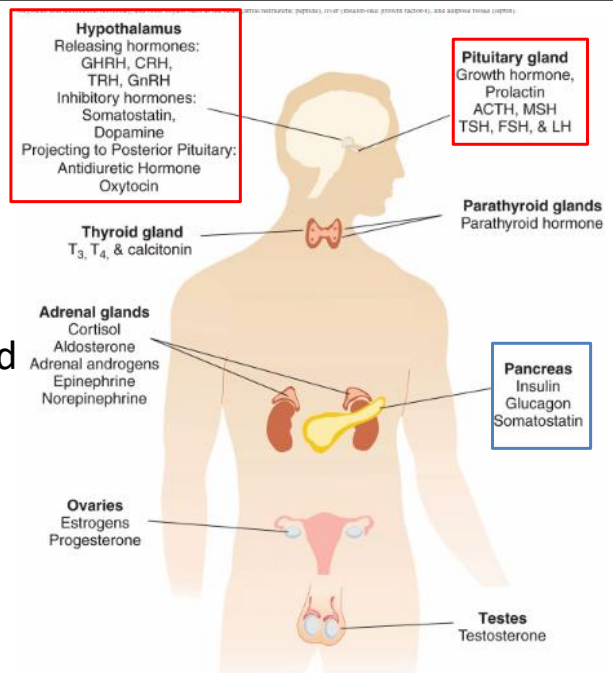
Outline

- 1- Endocrine system
- 2- Types of hormones
- 3- Regulation of hormone secretion
- 4- Pineal gland
- 5- Structure, functions and Abnormalities of the pituitary gland
- 6- Management strategies

Endocrine System

Endocrine system comprises:

- 1- Endocrine tissue of the pancreas.
- 2- Pituitary gland, hypothalamus and pineal gland.
- 3- Thyroid and parathyroid.
- 4- Adrenal
- 5- Gonads



Hormone

- Endocrine hormones are chemical compounds that are synthesized and released in small amounts into the systemic circulation to mediate a physiological function.
- Chemical structure:
 - ✓ **Steroids:**
Derivatives of cholesterol (hormones from gonads and adrenal).
Lipid soluble. Can cross the phospholipid bilayer, activate receptors inside the cells or turn genes on/off in the nucleus.

✓ **Non-steroid (amines or protein):**

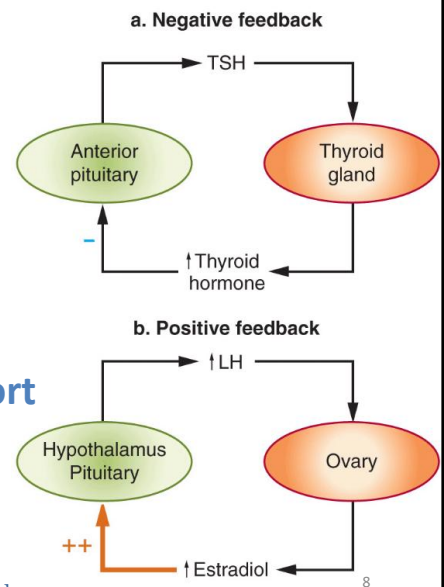
A- protein nature: Synthesized in rough ER as prohormones & packaged in vesicles till release (Insulin, glucagon).

Most hormones have protein nature. Hydrophilic in nature. Bind to cell surface receptors and activate secondary messengers to induce their physiological function.

B- Amines: derived from tyrosine (thyroid hormones and adrenaline/noradrenaline).

Regulation of hormone function

- ✓ **Regulation of hormone release:**
 - Negative feedback mechanism (predominant)
 - Positive feedback mechanism
- ✓ **Regulation of hormone receptors**
 - Down-regulation or up-regulation of receptors.
- ✓ **Others: duration, hormone release, transport and elimination.**



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- Transport: water-soluble hormones vs steroid & thyroid hormone.
- Plasma protein binding = reservoir and slows clearance from the plasma.
- Cyclical Variations: variations in hormone release pattern according to developmental stage or circadian clock. Example:
 - Sex hormones
 - Growth hormone
 - Cortisol

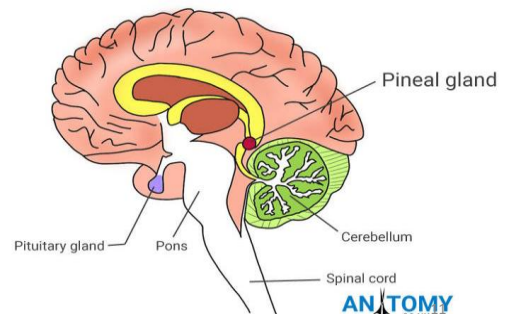
Pineal, Pituitary and Hypothalamus

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Pineal gland

- The pineal gland (aka, epiphysis): small gland beside the 3rd ventricle.
- Main secretory cells, pinealocytes: secrete melatonin (amine hormone; tryptophan/serotonin derivative).
- Melatonin release is stimulated by darkness and inhibited by daylight.
- Melatonin is important in regulating the circadian rhythm and sleep pattern.



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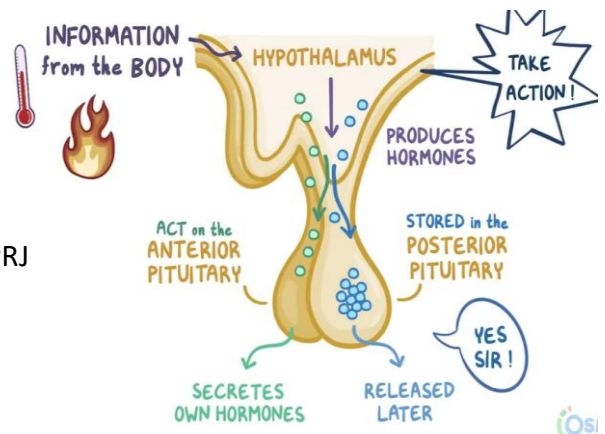
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Hypothalamus

“neuroendocrine organ”

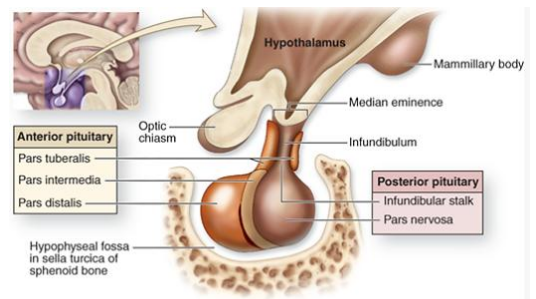
The hypothalamus has important role in several physiological functions including;

- 1- Temperature regulation
- 2- Neuroendocrine control of catecholamines, posterior pituitary gland (ADH, oxytocin) and anterior pituitary gland (via TRH, CRH, GnRH, PRJ and GHRH).
- 3- Regulate thirst, hunger and satiety.
- 4- Aid in controlling the circadian rhythm.



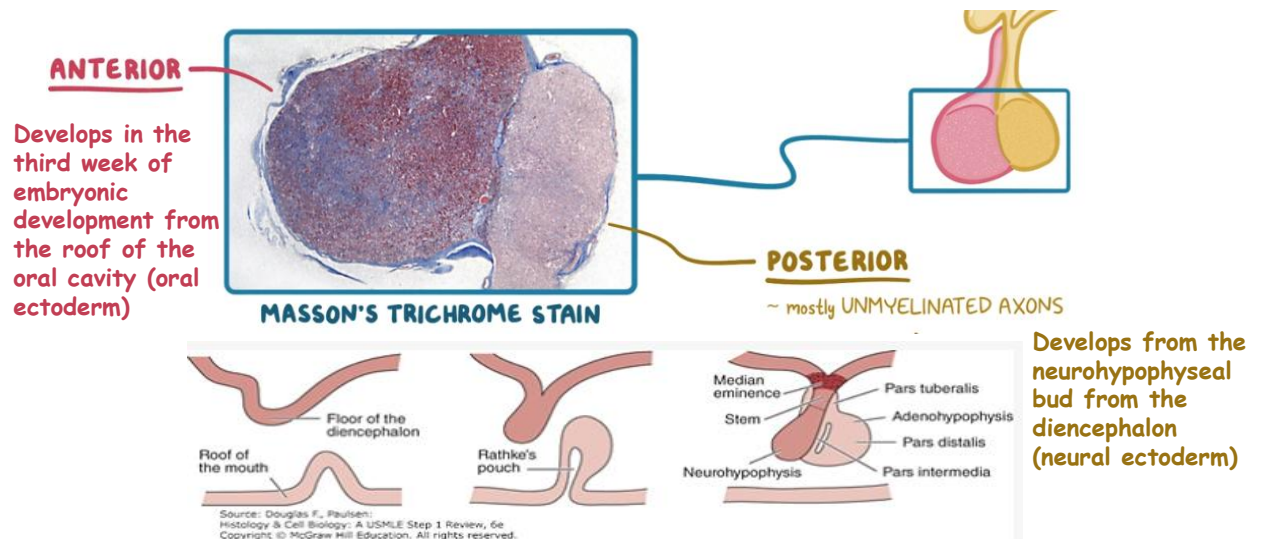
Pituitary

- Pituitary (*aka, hypophysis*) is a small pea-sized gland (0.5 gm) in the hypophyseal fossa.
- Composed of an adenohypophysis and neurohypophysis.
- Directly attached to the hypothalamus by an infundibular stalk.



Source: Anthony L. Mescher:
Junqueira's Basic Histology Text and Atlas, 16e
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Embryonic development



Histology

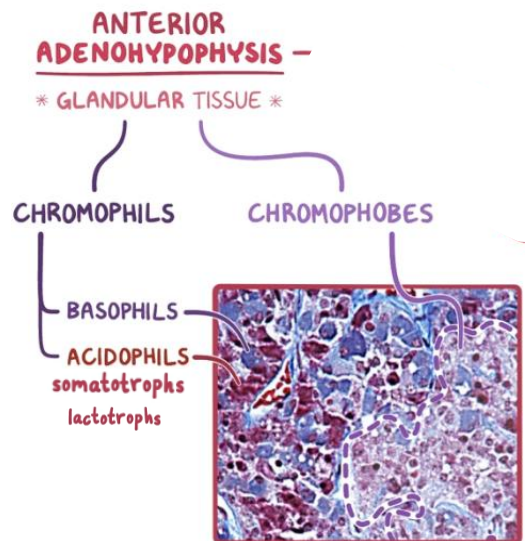
The adenohypophysis includes five distinct cell populations responsible for secreting six peptide hormones.

1- Chromophils (stain-loving cells, they stain darker in H & E staining)

a) Acidophils “stain dark red”

i) Somatotrophs, aka somatotropes, secrete somatotropin or growth hormone.

ii) Lactotrophs, aka lactotropes, produce prolactin (PRL),



b) Basophils “stains purplish blue; **B-FLAT**”

i) Gonadotrophs: produce follicle-stimulating hormone (FSH) & luteinizing hormone (LH).

ii) Corticotrophs, aka corticotrophs, produce adrenocorticotrophic hormone (ACTH)

iii) Thyrotrophs, aka thyrotrophs, produce thyroid-stimulating hormone (TSH)

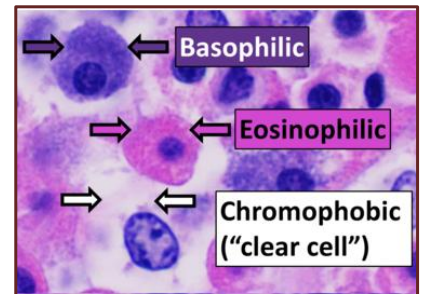
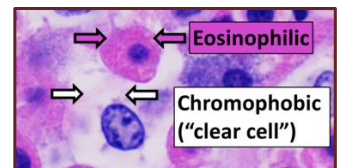


Image by Mikael Häggström, MD.

2- chromophobes: cells with lighter staining. They contain some secretory granules. Chromophobes include:

- I. melanotrophs that secrete melanocyte stimulating hormone
- II. stem cells that can differentiate into hormone-producing cells and older chromophil cells that lost their secretory function.
- III. The folliculo-stellate cells: have phagocytic and paracrine function (to maintain local homeostasis in the pituitary gland).



Neurohypophysis:

1- Unmyelinated axons.

Major population. Neurosecretory activity, they carry oxytocin and anti-diuretic hormone from the hypothalamus to nerve terminals to axonal distension, named **herring bodies**.

Herring bodies store and release the hormones. Released hormones are carried away by blood capillaries.

2- Blood capillaries

3- Pituicytes: glial cells that provide supportive function to the axons.

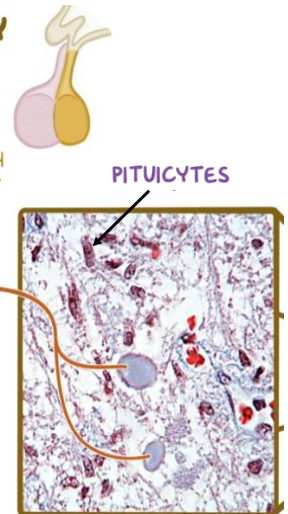
POSTERIOR PITUITARY

~ UNMYELINATED AXONS

NEUROSECRETORY
carry OXYTOCIN & ADH

HERRING BODIES

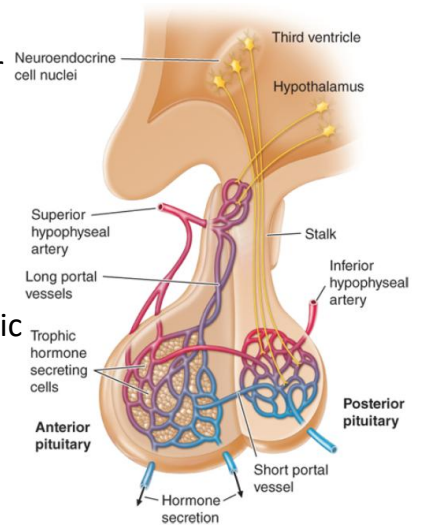
dilated nerve terminals
neurosecretory bodies
release
OXYTOCIN or ADH



Blood supply

Blood supply of the pituitary gland comes from the superior and inferior hypophyseal arteries (from carotid).

- Anterior pituitary: superior hypophyseal arteries form capillary network in the hypothalamus and drain to the anterior pituitary forming a “hypothalamic–hypophyseal portal system”. To allow transmission of hypothalamic peptides directly to the anterior pituitary without systemic dilution.
- The posterior pituitary gland, derived from neural tissue. Receives blood from the inferior hypophyseal artery. Directly innervated by hypothalamic neurons via the pituitary stalk



Source: Joseph Loscalzo, Anthony Fauci, Dennis Kasper, Stephen Hauser, Dan Longo, J. Larry Jameson: Harrison's Principles of Internal Medicine, 21e Copyright © McGraw Hill. All rights reserved.

Hypothalamic-Anterior Pituitary Gland

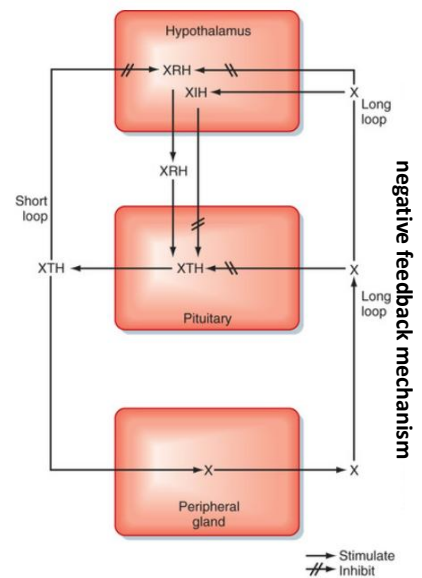
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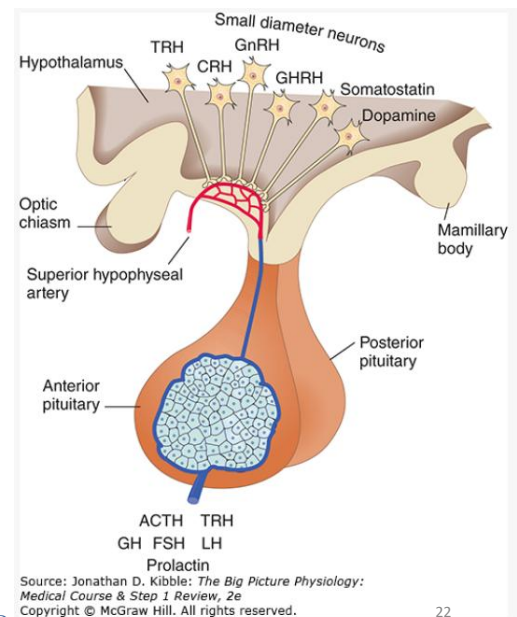
Endocrine system: three-level system

- This organization allow central control over endocrine function. Hypothalamus-pituitary-peripheral gland axis (except, prolactin).
 - 1- Neuroendocrine hypothalamic neurons (via releasing hormones).
 - 2- Anterior pituitary endocrine cells (via tropic hormones).
 - 3- Peripheral endocrine gland (specific hormone).

The neurovascular link (i.e, infundibular stalk) can be disrupted by physical trauma, or hypothalamic disease.



- Hypothalamus controls the anterior pituitary through releasing or inhibitory hormones via the “hypothalamic-hypophyseal portal system (aka, hypothalamic portal system)”.
- Hypothalamus releases:
 - **TRH**, thyrotropin-releasing hormone.
 - **CRH**, corticotropin-releasing hormone.
 - **GnRH**, gonadotropin-releasing hormone.
 - **GHRH**, growth hormone-releasing hormone.
 - **GH-inhibiting hormone** (Somatostatin).
 - **Prolactin-inhibiting hormone** (Dopamine)



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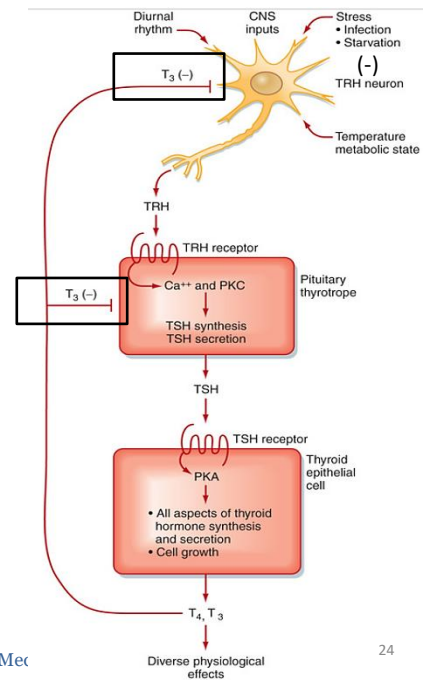
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- Hypothalamic hormones are released at certain intervals (not continuous and may follow the circadian rhythm).
- In response, the pituitary gland will release
 - Thyroid stimulating hormone (TSH)
 - Luteinizing hormone (LH)
 - Follicle stimulating hormone (FSH)
 - Adrenocorticotrophic hormone (ACTH)
 - Growth hormone
 - Prolactin

TRH & TSH (thyrotropin)

Synthesis, stimulation:

- TSH, glycoprotein hormone, synthesized by thyrotropes in response to thyrotropin-releasing hormone (TRH).
- TRH is released according to a diurnal rhythm (highest during overnight hours).
- TRH secretion is inhibited by stress, starvation, infection as well as thyroid hormones, T_3 and somatostatin (ss).



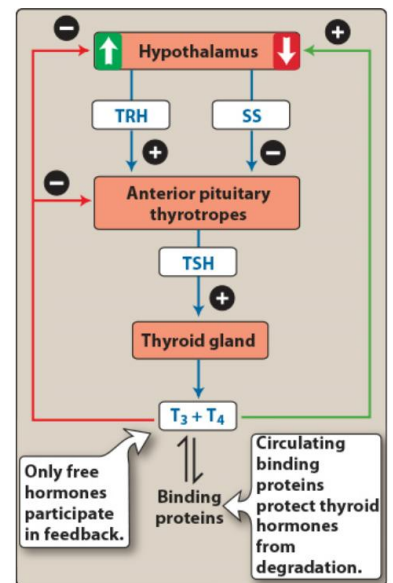
Regulation

- 1- T₃/T₄ decrease its secretion.
- 2- Cold temperature activate TRH and hence TSH secretion.

Function:

- 1- TSH stimulates synthesis and release of thyroid hormones (T₄ and T₃) via activating synthetic pathway (starting material: iodide and thyroglobulin)
- 2- Growth factor for cells of thyroid gland, i.e increase the size, number and increased secretory activity of thyroid cells

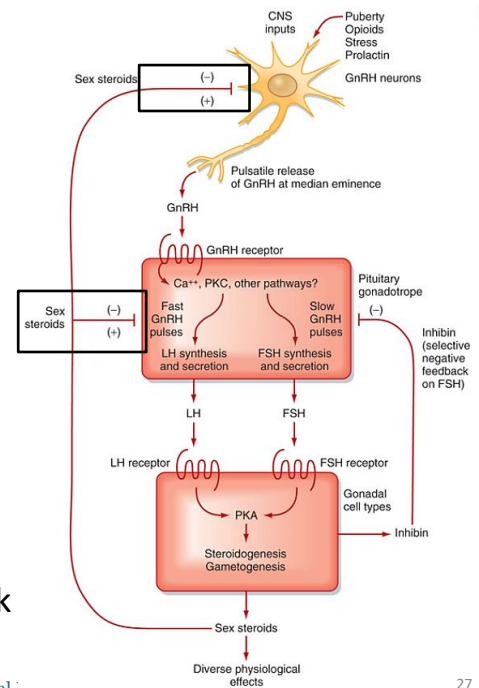
- T3(triiodothyronine) and T4 (thyroxine)
 - transported in blood bound to thyroxine-binding globulin
 - In the cell, T4 is enzymatically de-iodinated to **T3**, which enters the nucleus and attaches to specific receptors.
 - T3: Activation of receptors maintains the basal metabolism (energy consumption and weight), affect the heart rate and stimulate the response to sympathetic nervous system.



GnRH: FSH and LH

Synthesis, stimulation:

- GnRH is released in a pulsatile manner (or intervals).
 - High frequency GnRH release stimulates LH secretion,
 - Slower or low frequency GnRH release preferentially stimulate FSH release.
- Puberty upregulate the GnRH function.
- Sufficient sex steroids hormones will inhibit FSH release.
- In contrast, LH release is stimulated by high estrogen levels during ovulation (positive feedback mechanism).



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Function:

Gonadotropins promote testosterone secretion in males as well as estrogen and progesterone secretion in females.

In females:

- Follicle stimulating hormone stimulate the growth of the ovarian follicles in the ovaries and regulate the ovulatory cycle.
- Luteinizing hormone stimulates ovulation, release of progesterone from corpus luteum, steroid release from the ovaries.

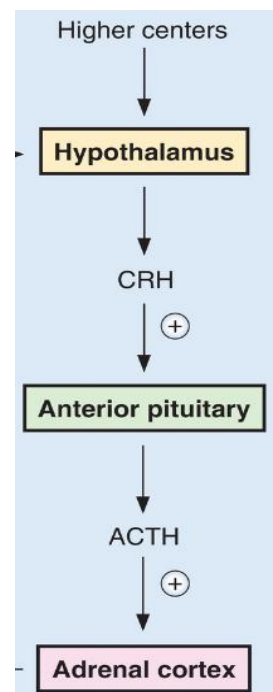
In males:

- FSH aids in spermatogenesis; while, Luteinizing hormone stimulates testosterone release.

CRH and ACTH

Synthesis:

- In response to corticotropin-releasing hormone (CRH), ACTH is synthesized by corticotropes (aka, POMC cells) from the prohormone precursor proopiomelanocortin (POMC).
- CRH and ACTH are released in pulses following the diurnal rhythm (highest levels in the early morning).
- ACTH stimulate the adrenal cortex.

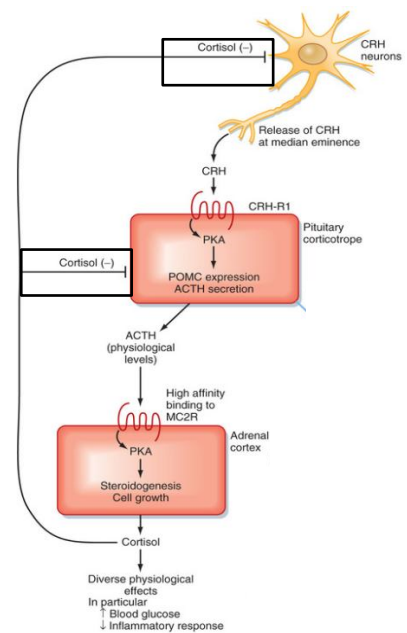


Function:

- ACTH binds to adrenal receptors in the adrenal cortex and stimulates the production and release of:
 - Glucocorticoids (like, cortisol) from the middle zona fasciculata
 - Mineralocorticoids (like, aldosterone) from the outer zona glomerulosa .
- Aldosterone is responsible for regulating salt and water metabolism.
- Glucocorticoids are involved in metabolism and response to stress.
- ACTH promotes growth and survival of cells in the adrenal cortex .
- ACTH plays a role in glucose metabolism and immune function.

Regulation:

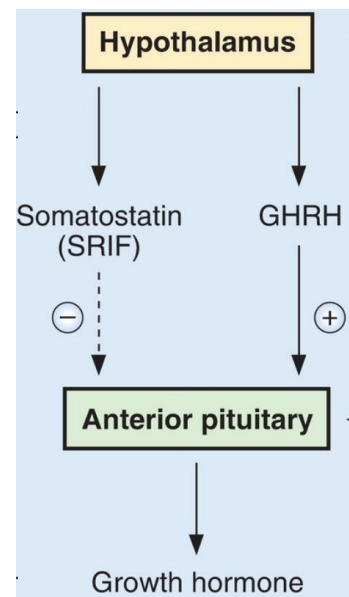
- ACTH is stimulated by psychological and physical stressors such as infection, hypoglycemia, surgery, and trauma.
- Cortisol inhibits CRH and ACTH secretion via a negative feedback mechanism.
- Hypothalamus can control the setpoint of HPA axis.



GHRH & Growth Hormone

Synthesis:

- GH (somatotropin) is synthesized by somatotrophs.
- Single-chain polypeptide, protein-anabolic hormone with structural similarity to prolactin.
- Release is controlled by the balance between stimulatory growth hormone-releasing hormone and inhibitory factor somatostatin. Both are released by the hypothalamus. GHRH has the more dominant function.



Endocrine axis: hypothalamic-pituitary-liver axis.

- GH is released in a pulsatile manner. Highest release occur at sleep and during exercise.
- GH stimulates production of insulin like growth factor-I (IGF, aka; somatomedin c) in the liver.
- IGF-1 is secreted continuously, has longer half-life and exhibits more stable concentrations in blood.

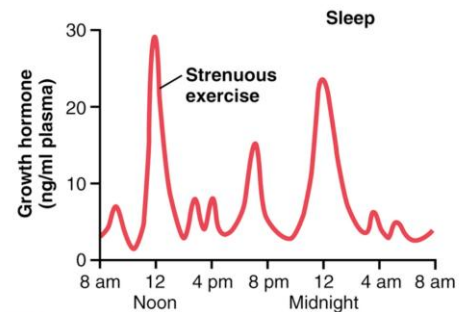
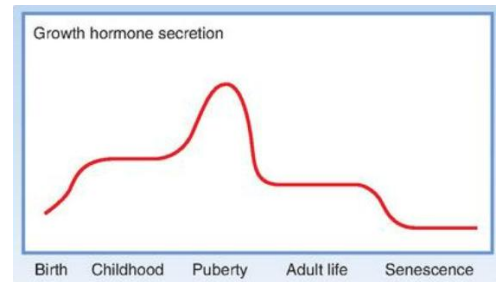


FIGURE 76-7. Typical variations in growth hormone secretion throughout the day, demonstrating the especially powerful effect of strenuous exercise and also the high rate of growth hormone secretion that occurs during the first few hours of deep sleep.



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Regulation of Growth hormone

- **Secretion is increased** by sleep, stress, hormones related to puberty, starvation, exercise & hypoglycemia.
- **Secretion is decreased** by somatostatin, somatomedins (IGF-1), obesity, hyperglycemia and aging.
 - Somatomedins inhibit GH through directly acting on the anterior pituitary.
 - GH also inhibits its secretion by stimulating somatostatin from the hypothalamus.
 - GHRH inhibits its own secretion from the hypothalamus.

Stimulate Growth Hormone Secretion

Decreased blood glucose
 Decreased blood free fatty acids
 Increased blood amino acids (arginine)
 Starvation or fasting, protein deficiency
 Trauma, stress, excitement
 Exercise
 Testosterone, estrogen
 Deep sleep (stages II and IV)
 Growth hormone-releasing hormone
 Ghrelin

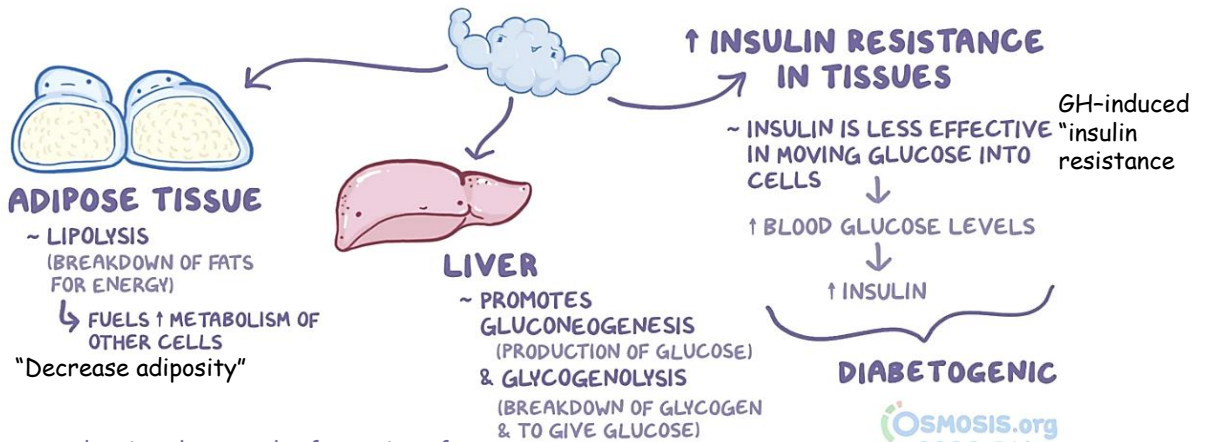
Inhibit Growth Hormone Secretion

Increased blood glucose
 Increased blood free fatty acids
 Aging
 Obesity
 Growth hormone inhibitory hormone (somatostatin)
 Growth hormone (exogenous)
 Somatomedins (insulin-like growth factors)

Functions of Growth hormone

DIRECT EFFECTS

~ STIMULATION OF CELLULAR METABOLISM



In the muscles, it enhances the formation of lean body mass via increasing the amino acid uptake and protein synthesis.

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INDIRECT EFFECTS



STIMULATES STEADY RELEASE
OF SOMATOMEDIN C

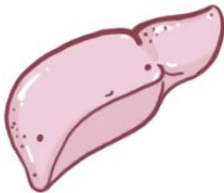


INSULIN-LIKE GROWTH FACTOR 1

- └ BINDS TO IGF-1 RECEPTORS/INSULIN RECEPTORS
- └ PROMOTES CELLULAR METABOLISM
- └ PREVENTS CELL DEATH
- └ ↑ RATE OF CELL DIVISION/DIFFERENTIATION

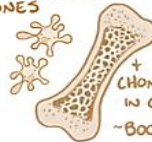
Promotes protein, RNA and DNA synthesis

Increases linear growth via action on chondrocytes and osteoblasts.



└ HELPS PROTEIN
PRODUCTION +
MUSCLE GROWTH

ACTIVITY OF
OSTEOBLASTS IN
BONES



+ CHONDROCYTES
IN CARTILAGE
~BOOSTS GROWTH



OSMOSIS.org

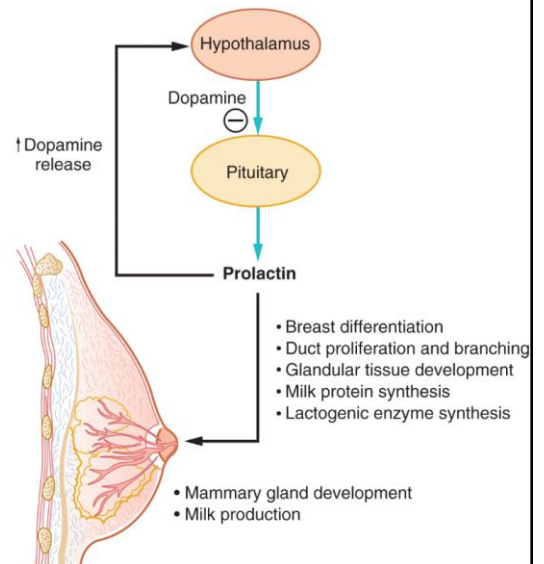
Prolactin

Synthesis:

Prolactin is synthesized by lactotrophs in response to thyrotropin-releasing hormone (TRH).

Responsible for lactogenesis.

Act directly on nonendocrine cells (breast) to induce trophic .



Factors that Increase Prolactin Secretion

Estrogen (pregnancy)
Breast-feeding
Sleep
Stress
TRH
Dopamine antagonists

Factors that Decrease Prolactin Secretion

Dopamine
Bromocriptine (dopamine agonist)
Somatostatin
Prolactin (by negative feedback)

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Prolactin (*cont.*)

Function:

- ❖ Stimulates mammary gland development and milk production.
- ❖ Inhibit ovulation and spermatogenesis via decreased GnRH.
- ❖ With the aid of estrogen signaling, prolactin stimulates the development of breast and mammary glands in females.
- ❖ Inhibits ovulation by decreasing synthesis and release of gonadotropin-releasing hormone (GnRH). Inhibits spermatogenesis (by decreasing GnRH).
- ❖ Prolactin modulates reproductive and parental behavior.

Pathophysiology of prolactin:

- ❖ Prolactin deficiency can lead to the disruption of milk production in the breast.
- ❖ Prolactin abundance can lead to causes galactorrhea (milk secretion from the breast in non-lactating females), decreased libido and decreased fertility because it inhibits GnRH secretion.

Melanocortin

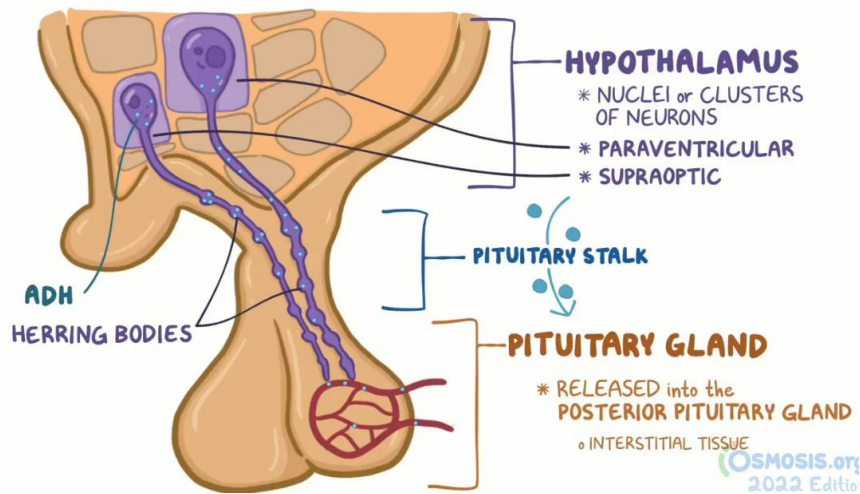
Synthesis and function:

Under physiological conditions, small amounts of MSH is produced by the melanotrophs (from POMC) following dopamine stimulation.

Melanocortin act on skin melanocytes to stimulate the melanin synthesis, i.e, important for skin pigmentation.

Decreased cortisol production (Addison's): supraphysiological levels of ACTH will increase melanin release leading to skin darkening.

Posterior Pituitary Gland



Anti-diuretic hormone

Synthesis, regulation

ADH (vasopressin) originate from the hypothalamic supraoptic nuclei.

In response to nerve stimulation, ADH is released in the nerve endings by exocytosis. Acts on the collecting ducts in the nephrons.

Serum osmolarity is the main regulator for ADH release. Factors that increase the serum osmolarity or decrease blood volume (bleeding, nausea) as well as low blood pressure will stimulate ADH secretion.

Factors that Increase ADH Secretion

↑ serum osmolarity
Volume contraction
Pain
Nausea (powerful stimulant)
Hypoglycemia
Nicotine, opiates, antineoplastic drugs

Factors that Decrease ADH Secretion

↓ serum osmolarity
Ethanol
α-Agonists
ANP

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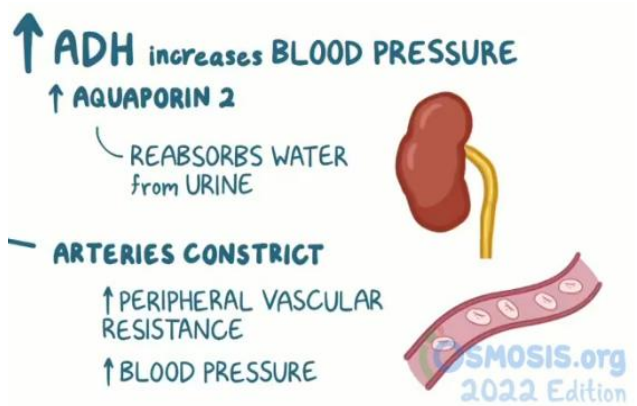
Function

1- In low concentration:

Antidiuretic effect. It decrease water excretion by the kidneys.

2- In higher concentration:

Vasoconstriction. ADH is a potent vasoconstrictor for arterioles leading to increased arterial pressure.



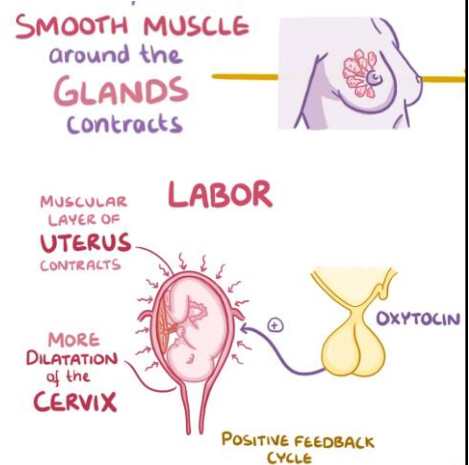
Oxytocin

Synthesis, regulation

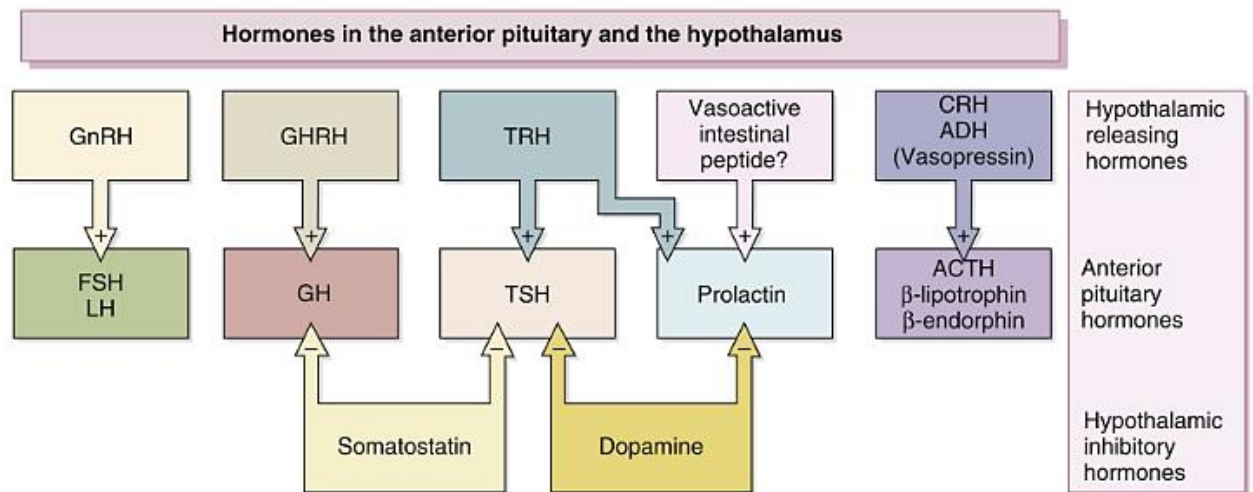
From the paraventricular nuclei of the hypothalamus. Release is stimulated by the suckling action during lactation and during labor (via a positive feedback mechanism).

Function

- ❖ Facilitate milk ejection from the mammary gland.
- ❖ Stimulate uterine contraction and cervical dilation to facilitate labor (positive feedback mechanism).
- ❖ Promotes maternal and social bonding behavior.



Summary for hypothalamic control



Endocrine Disorders

Abnormalities of Pituitary Gland

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Primary vs Secondary vs Tertiary endocrine Disorders



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Hypopituitarism

Hypopituitarism decreased secretion of one or more pituitary hormones. **Panhypopituitarism** decreased secretion of all pituitary hormones. They can occur:

1- Congenital

2- Secondary to

- head trauma or thrombus in pituitary blood vessels,
- radiation and/or autoimmune disorders/infections.
- **pituitary tumors** that compress the pituitary gland or destroys its ability to secrete hormones.

Adult hypopituitarism

- ACTH deficiency leading to adrenal insufficiency.
- TSH deficiency leading to hypothyroidism.
- Gonadotropin deficiency leading to hypogonadism and declining sexual functions.
- Decreased growth hormone secretion leading to decrease fat mobilization, weaker muscles and lower bone density.
- ADH deficiency leading to diabetes insipidus.
- Melanocortin deficiency leading to abnormal pallor.

Collectively leading to obesity, decreased libido and lethargy.

Dwarfism

A) Panhypopituitarism during childhood. Prevalence: around 70 %

- Decreased rate of body development. However, the different parts of the body will develop in appropriate proportions to each other.
- In panhypopituitarism, patients will lack adult sexual functions due to gonadotropins deficiency.

B) Due to deficiency of growth hormone. Around 30 % of cases.

Those individuals will have slower growth rate, but they attain sexual maturity and can reproduce.

C) Deficiency of somatomedin C. “short stature and slower growth”

- Dental complications of hypopituitarism:
 - Maxilla and mandible are smaller than the normal
 - delayed exfoliation of primary teeth
 - delayed eruption of permanent teeth.
 - shorter roots and crowding of permanent teeth.



50-y-old female with short stature

[Dental Abnormalities in Pituitary Dwarfism: A Case Report and Review of the Literature \(hindawi.com\).](#)
[Oral manifestations in growth hormone disorders - PMC \(nih.gov\)](#)

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Management of GH Deficiency

Examples: Synthetic human somatotropin (Humatrope®)

Indications: GH deficiency in children with growth retardation. May be used as a GH replacement in adults with confirmed deficiency.

Adverse effects: Increased risk of diabetes, edema and pain at injection site.

Hyperpituitarism

Attributed to the hypersecretion of the pituitary hormones.

Causes:

- 1- Pituitary adenoma (common)
- 2- Congenital due to genetic mutation in single cell yielding hormone-secreting tumor.

Clinical presentation:

- 1- Adulthood: Acromegaly.
- 2- Childhood: Gigantism.

Acromegaly

Most cases are secondary to GH-secreting pituitary adenoma. Clinical manifestation: multiple due to high levels of IGF-1.

Presentation include:

- 1- Organomegaly: heart, tongue (macroglossia), kidney and vocal cords
- 2- Joint pain and hunched back (kyphosis due to changes in the vertebrae).
- 3- onset of insulin resistance



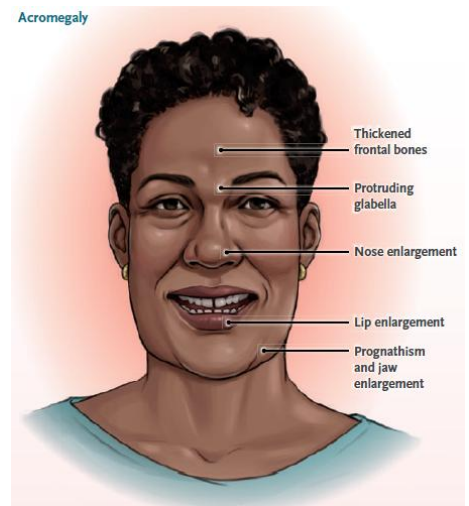
Figure 18.1 The clinical appearance of acromegaly. The facial features are coarse and mandibular overgrowth has lead to prognathism.



Figure 18.2 The appearance of the hands in the case of acromegaly. The acromegalic right hand is larger with expanded soft tissues and thickened, stubby fingers.

4- orofacial:

Thick lips, mandibular overgrowth and protrudes forward (prognathism), jaw thickening, maxillary widening, drifting of teeth, and malocclusion. In addition, enlargement of forehead bones and nose.



Gigantism

Less common disorder, mostly males.

Presentation:

Rapid growth of all body tissues that can occur before adolescence.

Orofacial: macroglossia, mandibular protrusion, longer roots, and extra deposition of cementum in the teeth (hypercementosis)

Hyperglycemia that can lead to destruction of beta cells and diabetes.



Figure 18.3 The worlds tallest ever recorded man, Robert Wadlow, was diagnosed with from gigantism as a consequence of pituitary hyperplasia. He stood at 2.27m/8ft 11in and wore a size 37 shoe.

Management of GH Over-production

A) Surgical

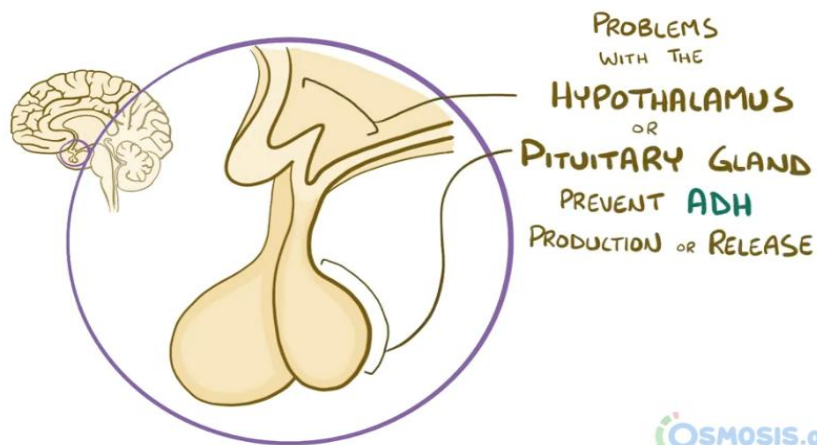
B) Pharmacological: Somatostatin is a growth hormone-inhibiting hormone

Example: Octreotide (Sandostatin®) and Lanreotide (Somatuline Depot®)

Indication: treatment of acromegaly

Adverse effects: GI symptoms (abdominal pain, diarrhea, nausea and vomiting). Bradycardia and orthostatic hypotension may also occur.

CENTRAL DIABETES INSIPIDUS



Etiology: damage to hypothalamus/pituitary gland, congenital defect leading to decreased ADH level.

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Management of ADH Deficiency

Example: Vasopressin (Vasopressin®) & Desmopressin (DDAVP®): the synthetic analogue.

Indication: Management of central diabetes insipidus.

Adverse effect: Desmopressin may cause xerostomia. Toxicity is associated with hyponatremia.

Syndrome of Inappropriate Anti-diuretic Hormone Secretion “SIADH”

Etiology:

Increased ADH release. Secondary to:

- 1- CNS disorders, head trauma, injury = damage for pituitary.
- 2- Ectopic production of ADH: in lung malignancies.
- 3- Drugs like carbamazepine (anticonvulsant), cyclophosphamide (chemotherapy) .

Pathophysiology

Characterized by excessive free water retention with continued urinary Na^+ excretion = hyponatremia and expansion of the plasma volume.

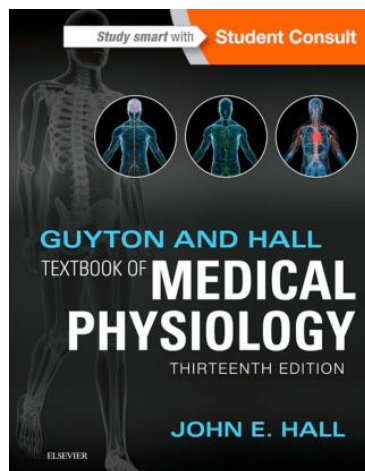
Symptoms:

Vary according to severity. Hyponatremia, fatigue, headache, dizziness, nausea, muscle cramps. Can lead to cerebral edema, seizures, coma.

Treatment:

Fluid restriction (first line), salt tablets and ADH antagonists

Resources



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